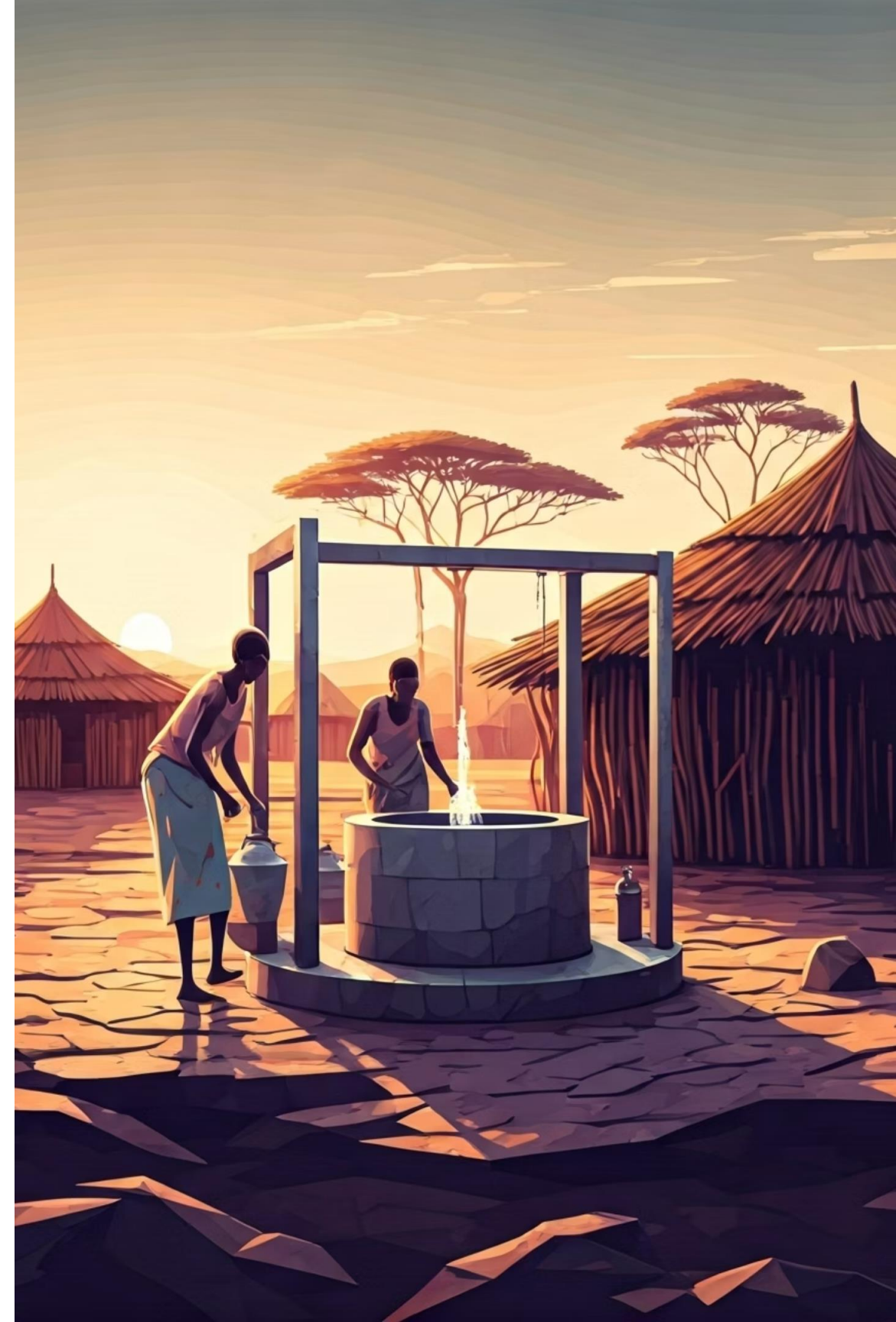


Predicting Water Well Functionality for Tanzania

Analysis By: Kodi Norman Mwapea, Ahjin Analytics

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The Challenge

70.5 Million

Tanzania's population in 2025

24-25 Million

**People relying on wells as
primary water source**

1 in 3

Tanzanians depend on well water daily

Access to safe and reliable water remains a daily uncertainty for millions. Wells are lifelines supporting health, livelihoods, and dignity—yet many fail within years due to poor construction, environmental stress, and limited maintenance.



The Human Cost of Well Failures

→ Health Impact

Communities exposed to waterborne diseases when wells fail

→ Economic Burden

Households incur losses securing alternative water sources

→ Educational Barriers

Women and children spend time collecting water instead of learning

→ Investment Waste

Donor and government funds lost when wells collapse prematurely



The burden of water collection disproportionately affects women and children, limiting educational opportunities and economic participation.

Our Data-Driven Solution

01

Predictive Modeling

Classify wells as functional, needs repair, or non-functional using machine learning

02

Proactive Maintenance

Technical teams prioritize wells at risk of failure, preventing downtime

03

Smart Investments

Direct resources to areas where wells are most vulnerable

04

Community Resilience

Ensure reliable access improves health, education, and economic participation





Total
records

Data columns

75%

Dataset Overview

59,400

Total Records

Water points across Tanzania

41

Data Columns

Comprehensive attributes

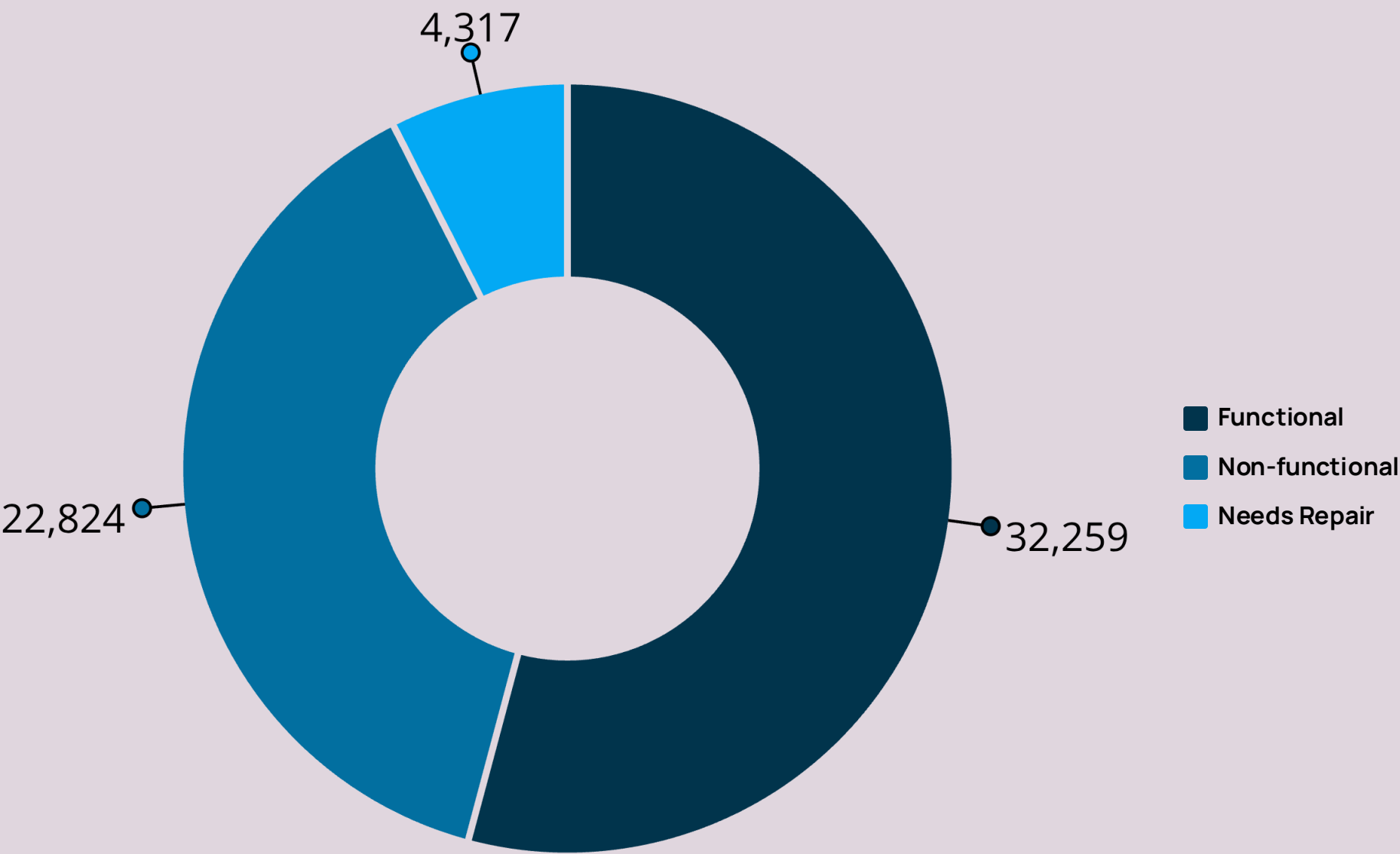
11.8

Years Covered

2001-2013 timeframe

Data sourced from Taarifa and Tanzania Ministry of Water, covering well location, construction type, water quality, management practices, and operational status. This comprehensive dataset provides rich foundation for predictive modeling.

Well Status Distribution



Key Factors Affecting Well Performance



Technology Type

Motorpumps (88.9%) and rope pumps (78.4%) show highest functionality, while older technologies like India Mark III (27.0%) fail more often.



Water Quality

Soft water points perform well (58.5%), while contaminated sources like milky (38.7%) and salty water (42.3%) show high failure rates.



Well Age

New wells (70.3%) vastly outperform old ones (32.6%), highlighting the critical need for infrastructure renewal.



Payment Systems

Wells with regular payments (annual 76.3%, per bucket 69.7%) maintain higher functionality than "never pay" systems (42.7%).

Regional Performance Disparities

Top Performing Regions



Iringa: 78.2%

Highest functional rate



Arusha: 69.7%

Strong performance



Lake Nyasa Basin: 74.8%

Best basin performance

Underperforming Regions



Mwanza: 27.8%

Needs urgent intervention



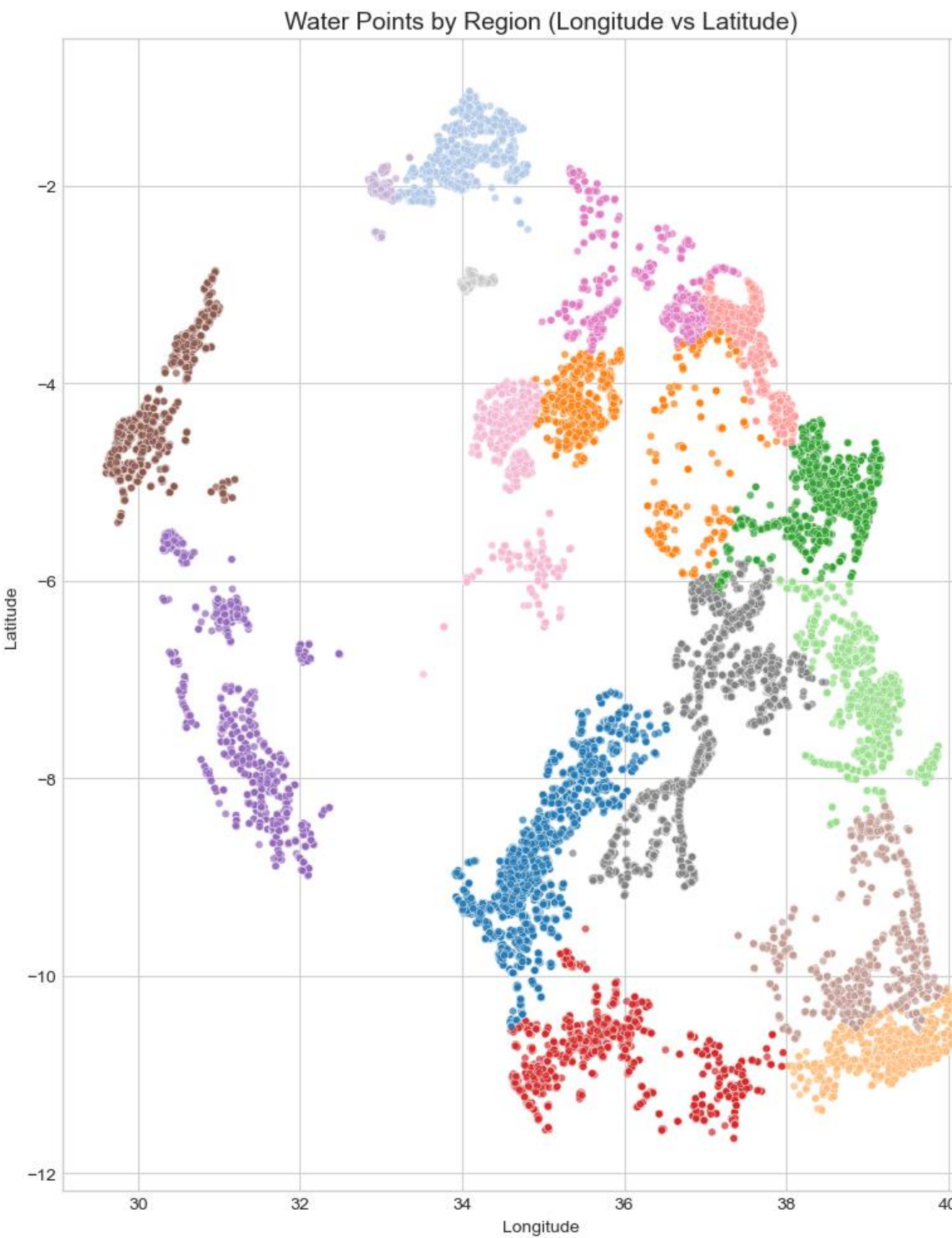
Lindi: 29.8%

Critical maintenance gaps



Lake Victoria Basin: 40.4%

Below average performance



Machine Learning Model Results



XGBoost

With 86.77% accuracy, XGBoost with class weights and GridSearchCV was the best est overall in detecting non-functional wells.



82% Recall Rate

The model successfully identified 82% of non-functional wells across all top models, crucial for preventive maintenance.



Balanced Performance

F1-score of 0.86 demonstrates strong balance between precision and recall for practical deployment.

Strategic Recommendations



Targeted Maintenance

Prioritize repairs for older infrastructure and high-failure regions like Mwanza and Lindi



Strengthen Community Management

Train Village Water Committees and user-groups to improve maintenance and oversight



Invest in Proven Technologies

Favor gravity-fed systems, motorpumps, and rope pumps while upgrading high-failure technologies



Deploy Predictive Model

Implement XGBoost classifier to proactively identify wells at risk of failure

From building wells to building resilience—ensuring every investment delivers lasting impact for Tanzanian communities.